

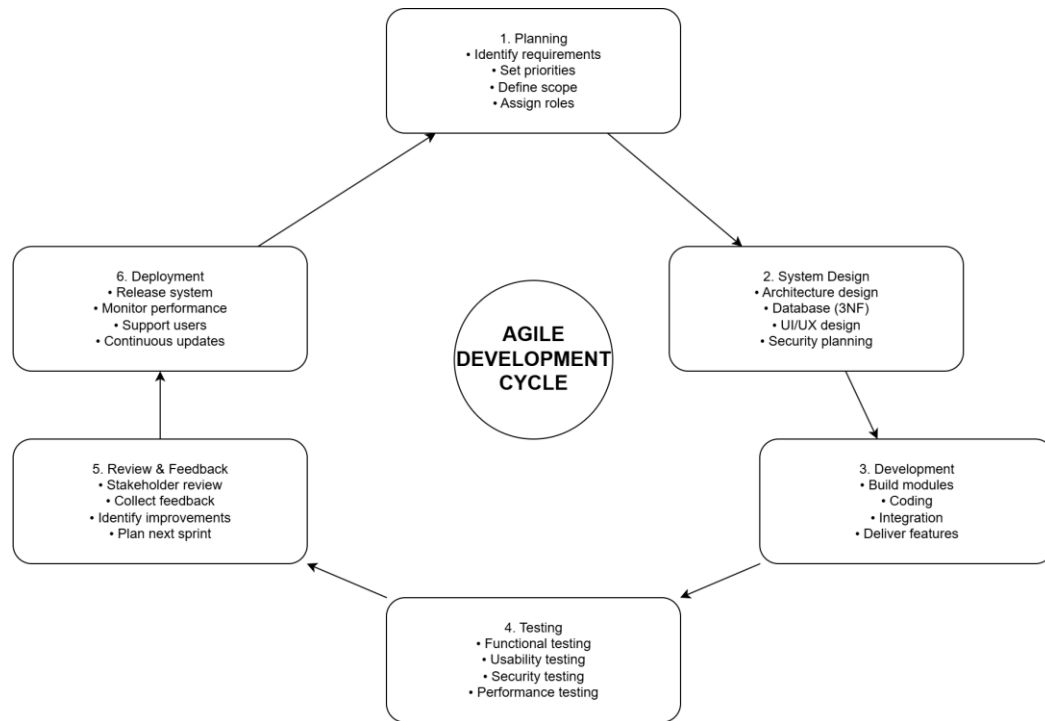
5. System Development Methodologies

Agile methodology can be viewed as the best methodology for developing the Digital Prescription and Medication Tracking Management System (DPMTMS) of the Bundoora Family Clinic. Even though all the necessary requirements for DPMTMS have been identified (digital prescriptions, medication tracking, inventory management, access control based on roles and artificial intelligence for decision making), the use of the Agile methodology will provide flexibility in improving and implementing the additional functions during the process. Given the fact that only six months will be needed for developing this system, the iterative approach of Agile will enable to receive feedback from stakeholders at early stages and to minimize the risk. The advantages of the Agile methodology will consist in high attention to collaboration, customer satisfaction and flexibility which will help in dealing with unexpected problems (encryption of medical data, for instance).

To provide a clear context for the chosen methodology, the core project parameters are summarized in the overview below:

Relevant Data	Goal
Digital prescriptions, inventory tracking, AI decision support, RBAC, and automated notifications.	A secure, user-friendly, and accessible web-based DPMTMS.
Resource Required	Time Line
Project Manager, DB Admin, UI Designer, Programmers, QA Tester, and continuous stakeholder feedback.	Six-month period

Agile process model works in an iterative and incremental fashion and includes various phases such as planning, designing, developing, testing, deploying and support review. The entire process model is done several times in short sprints and is a betterment in terms of the software that is developed. The following diagram shows the life cycle of Agile model.



1. **Planning:** At this stage of the Agile SDLC process, the clinic stakeholders and project manager discuss priorities and scope of the project. It is also vital to define roles and responsibilities of the development, database, and design teams.
2. **System Design:** The significance of this stage is in making sure the software meets functional requirements, which include 3NF database design and MVC architecture. It is vital to make the right choice of efficient development frameworks and safe programming environment to manage health care data.
3. **Development:** This stage is marked with programmers and web developers initiating the process of implementation of the system. It is the longest and most collaborative stage. As a result, it leads to the creation of the first draft of the product that consists of the user management module and doctor module.
4. **Testing:** This stage is marked with the QA team performing testing of all the system features for functionality, usability, and security. The QA team tests every feature for its functionality and adheres to strict standards of health care data.
5. **Review and feedback:** After the testing stage, the stakeholders provide feedback, which is going to be used at the next stage.
6. **Deployment:** While doing deployment for the project, certain risks and scope creep situations might occur and updates might become necessary. In this phase, the team makes sure that there is seamless user experience.

Justification:

As all the stakeholders of the clinic (doctors, pharmacists, and administrative staff) will be present during the entire process of the project, Agile is the best methodology that can be used. The development of the software can take place along with getting inputs from the stakeholders and the requirement for the digital prescription system, AI drug interaction checker system, and inventory management system can be refined along the process.

The Waterfall Model and the Iterative Models take more time and money to alter due to their sequential nature or long cycle time. However, the Agile model avoids risks and creates software fast which satisfies the customers as well.

Reference:

Ilieva, et al. (2007), "Analyses of an agile methodology implementation". [Accessed at: https://research.uni-sofia.bg/xmlui/bitstream/handle/10506/11/SPPI04_31_ilievas.pdf?sequence=1]